

R CODE LAB EXPERIMENT

Set 1: Monthly Sales Data (R Code)

R

Load libraries

```
library(ggplot2)
```

```
library(plotly)
```

Dataset

```
sales <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000),  
  Advertising = c(2000, 2500, 3000, 2800, 3200)  
)
```

1. Line Chart

```
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Monthly Sales Trend",  
    x = "Month",  
    y = "Sales ($)"  
  )
```

2. Bar Chart

```
ggplot(sales, aes(x = Month, y = Sales)) +  
  geom_bar(stat = "identity", fill = "skyblue") +
```

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```
labs(  
  title = "Monthly Sales",  
  x = "Month",  
  y = "Sales ($)"  
)
```

3. Scatter Plot

```
ggplot(sales, aes(x = Advertising, y = Sales)) +  
  geom_point(size = 4, color = "darkgreen") +  
  labs(  
    title = "Advertising Budget vs Sales",  
    x = "Advertising Budget ($)",  
    y = "Sales ($)"  
  )
```

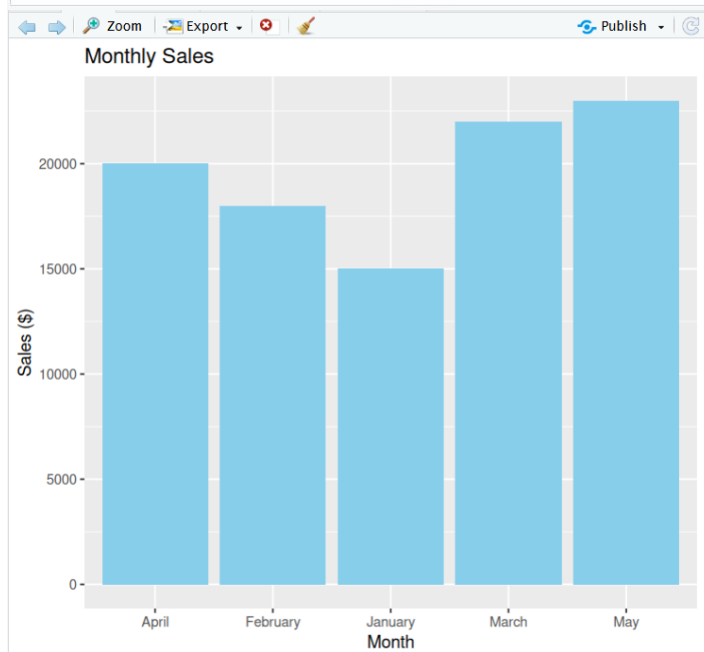
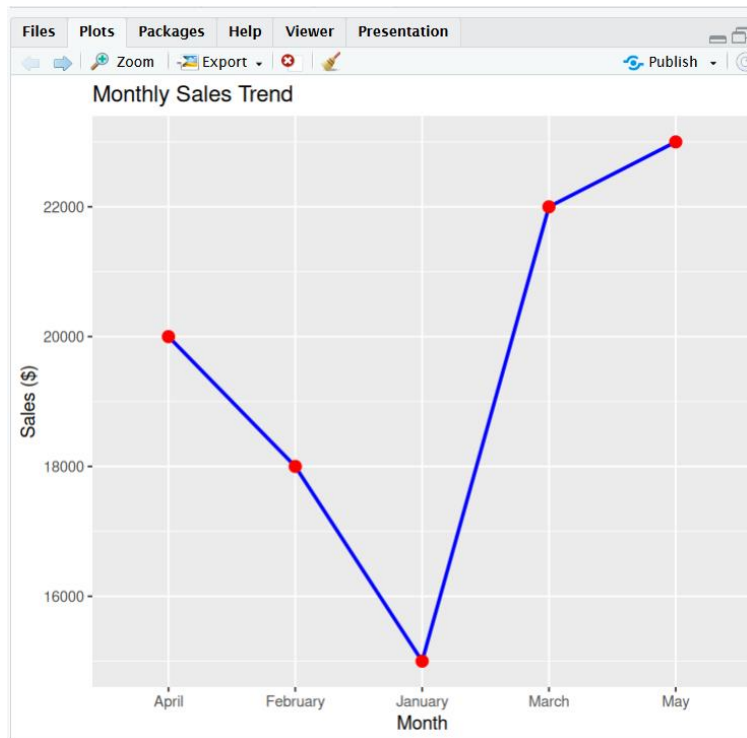
Insight:

Higher advertising budget generally results in higher sales.

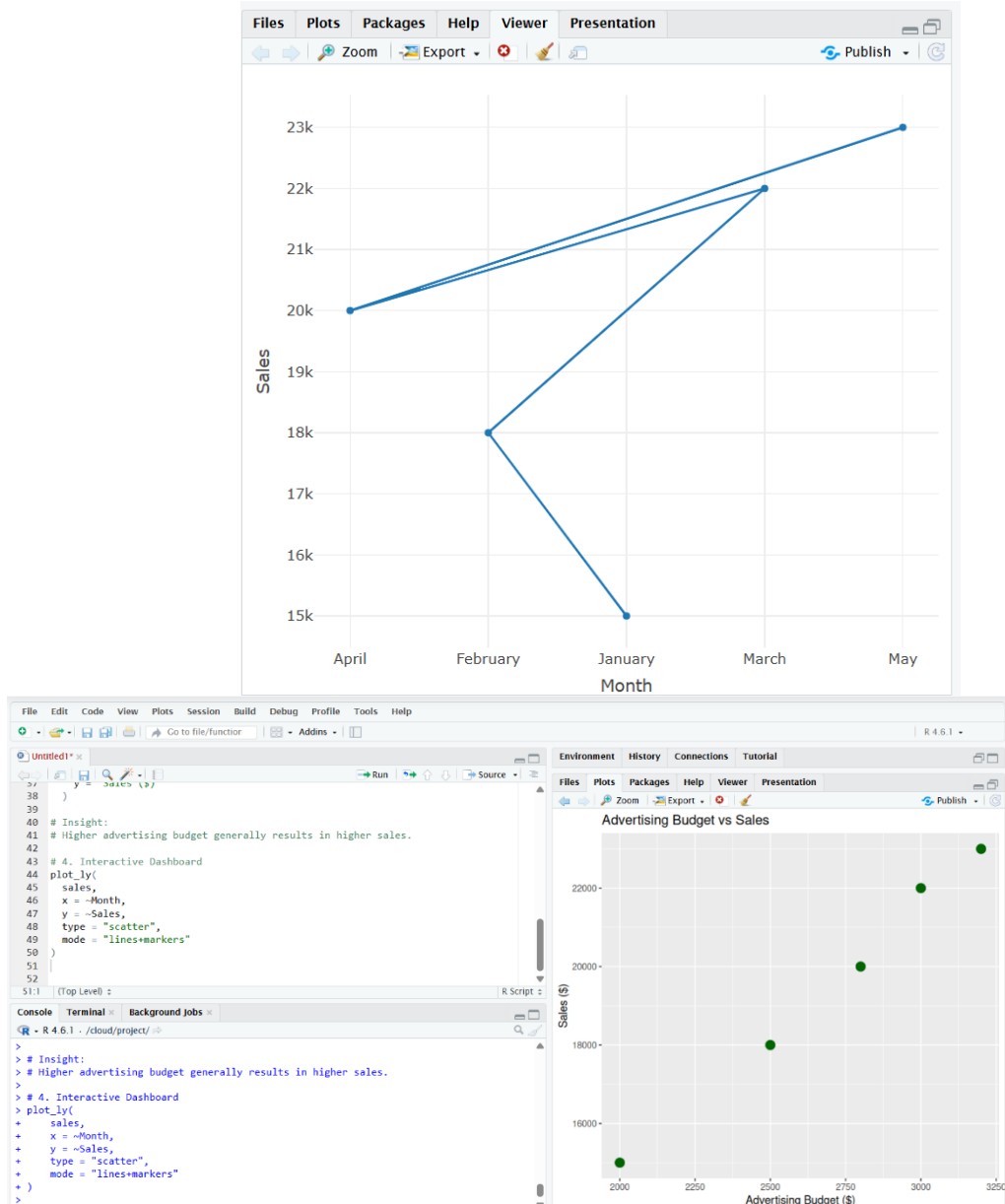
4. Interactive Dashboard

```
plot_ly(  
  sales,  
  x = ~Month,  
  y = ~Sales,  
  type = "scatter",  
  mode = "lines+markers"
```

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Set 2: Customer Feedback Analysis (R Code)

R

Load libraries

library(ggplot2)

library(wordcloud)

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Dataset

```
customer <- data.frame(  
  Customer_ID = c(1,2,3,4,5),  
  Age = c(25,30,35,28,40),  
  Satisfaction_Score = c(4,5,3,4,5)  
)
```

-----

1. Histogram for Customer Age Distribution

-----

```
ggplot(customer, aes(x = Age)) +  
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +  
  labs(  
    title = "Distribution of Customer Ages",  
    x = "Age",  
    y = "Frequency"  
  )
```

-----

2. Pie Chart for Satisfaction Scores

-----

```
score_count <- table(customer$Satisfaction_Score)  
  
pie(  
  score_count,  
  labels = names(score_count),  
  main = "Customer Satisfaction Score Distribution"
```

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)

3. Stacked Bar Chart by Age Group

```
customer$Age_Group <- c(
```

```
  "20-30",
```

```
  "20-30",
```

```
  "31-40",
```

```
  "20-30",
```

```
  "31-40"
```

```
)
```

```
ggplot(customer,
```

```
  aes(
```

```
    x = Age_Group,
```

```
    fill = factor(Satisfaction_Score)
```

```
  )) +
```

```
geom_bar() +
```

```
labs(
```

```
  title = "Customer Satisfaction by Age Group",
```

```
  x = "Age Group",
```

```
  y = "Number of Customers",
```

```
  fill = "Satisfaction Score"
```

```
)
```

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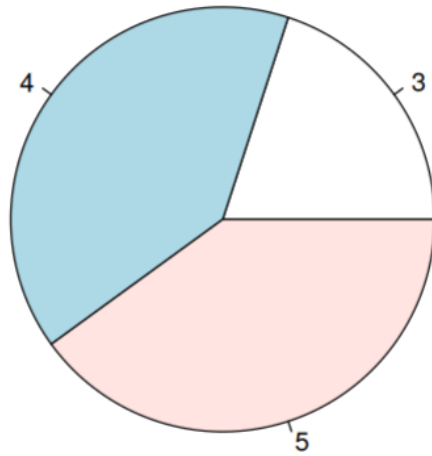
4. Word Cloud for Customer Feedback

```
feedback <- c(
  "good service",
  "excellent support",
  "fast delivery",
  "happy customer",
  "excellent experience",
  "good quality",
  "great support",
  "satisfied customer"
)

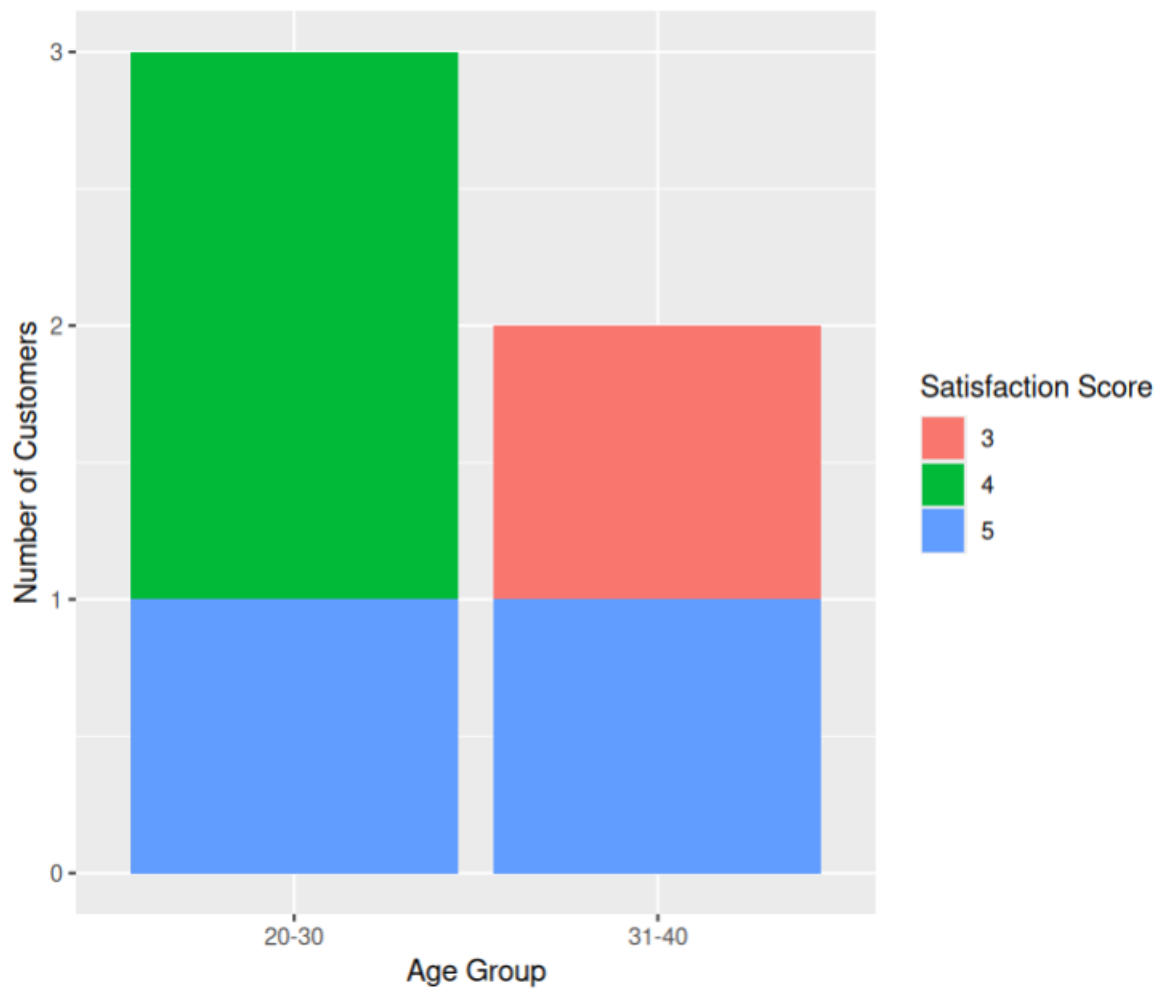
wordcloud(
  words = feedback,
  freq = rep(1, length(feedback)),
  scale = c(4,1)
)
```

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Customer Satisfaction Score Distribution



Customer Satisfaction by Age Group



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Set 3: Employee Performance Evaluation (R Code)

R

Load library

```
library(ggplot2)
```

Dataset

```
employee <- data.frame(  
  Employee_ID = c(1,2,3,4,5),  
  Department = c("Sales","HR","Marketing","Sales","HR"),  
  Years_of_Service = c(5,3,7,4,2),  
  Performance_Score = c(85,92,78,90,76)  
)
```

-----

1. Line Chart - Employee Performance Trend

-----

```
ggplot(employee,  
  aes(x = Employee_ID,  
    y = Performance_Score,  
    color = Department,  
    group = Department)) +  
  geom_line(linewidth = 1) +  
  geom_point(size = 3) +  
  labs(  
    title = "Employee Performance Trend",
```

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```
x = "Employee ID",  
y = "Performance Score",  
color = "Department"  
)
```

```
# -----
```

```
# 2. Bar Chart - Employees by Department
```

```
# -----
```

```
ggplot(employee,  
  aes(x = Department,  
      fill = Department)) +  
geom_bar() +  
labs(  
  title = "Employee Distribution by Department",  
  x = "Department",  
  y = "Number of Employees"  
)
```

```
# -----
```

```
# 3. Scatter Plot - Years of Service vs Performance
```

```
# -----
```

```
ggplot(employee,  
  aes(x = Years_of_Service,  
      y = Performance_Score,  
      color = Department)) +  
geom_point(size = 4) +  
labs(  
  title = "Years of Service vs Performance",  
  x = "Years of Service",  
  y = "Performance Score",  
  color = "Department"  
)
```

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```
title = "Years of Service vs Performance Score",  
x = "Years of Service",  
y = "Performance Score"  
)
```

```
# -----
```

```
# Insight:
```

```
# Employees with more years of service generally
```

```
# tend to have better performance scores,
```

```
# although exceptions exist.
```

```
# -----
```

```
# -----
```

```
# 4. Tableau Dashboard
```

```
# -----
```

```
# Import the dataset into Tableau.
```

```
# Create:
```

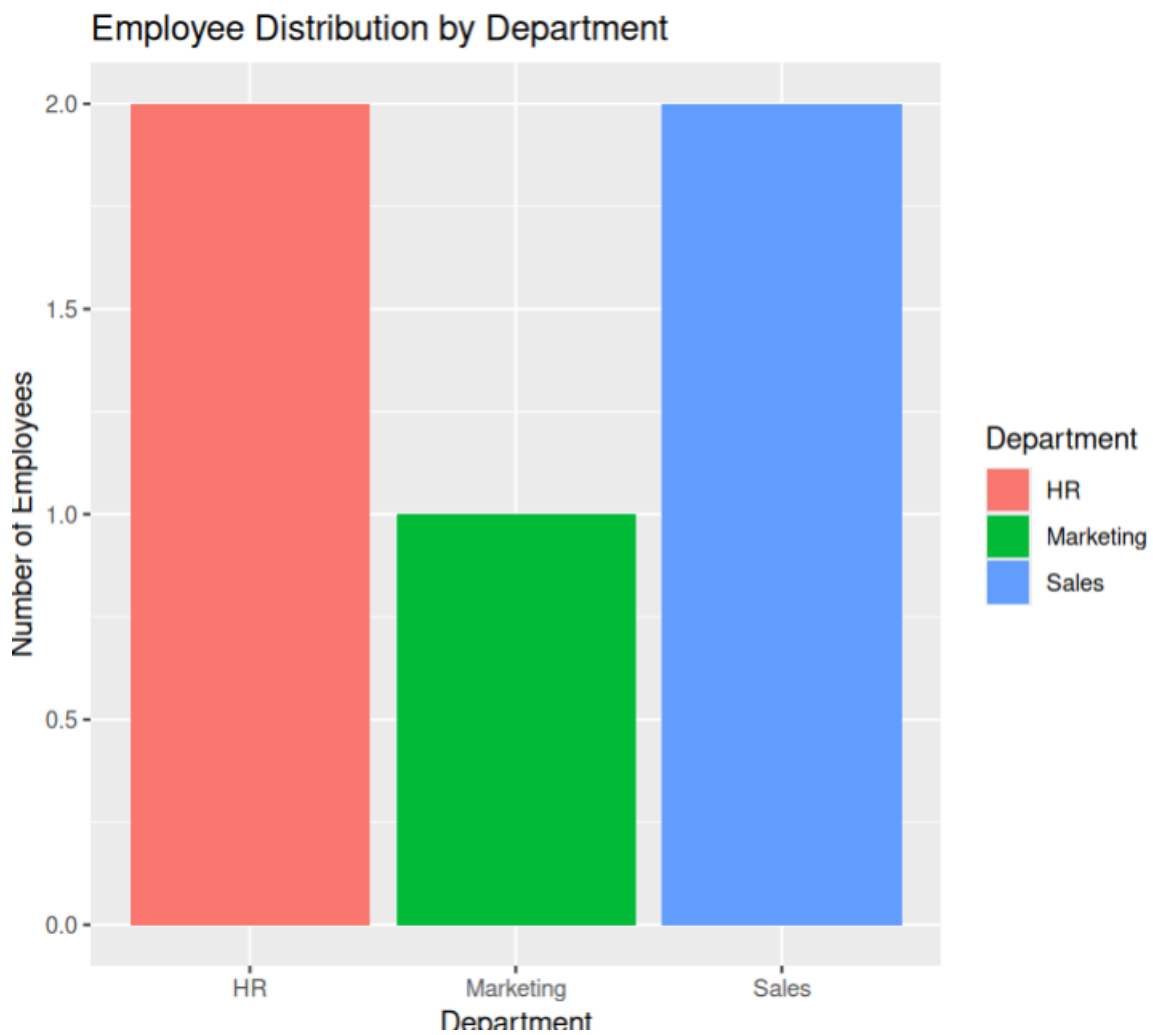
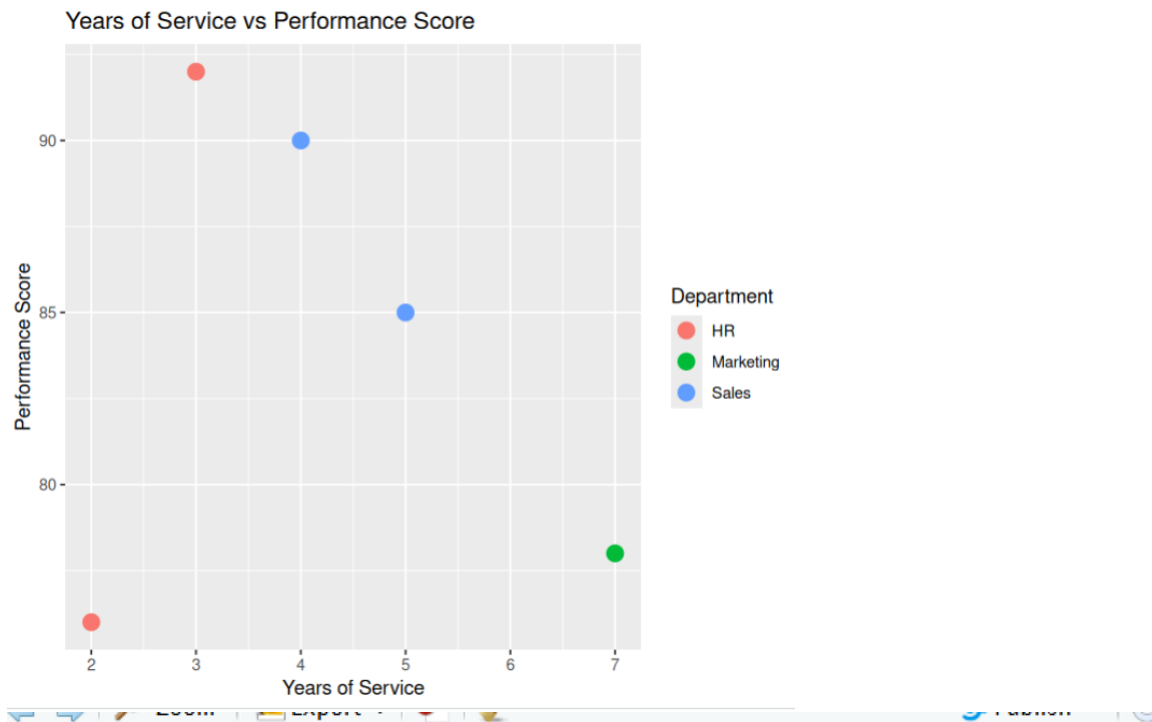
```
# Sheet 1 -> Line Chart
```

```
# Sheet 2 -> Department Bar Chart
```

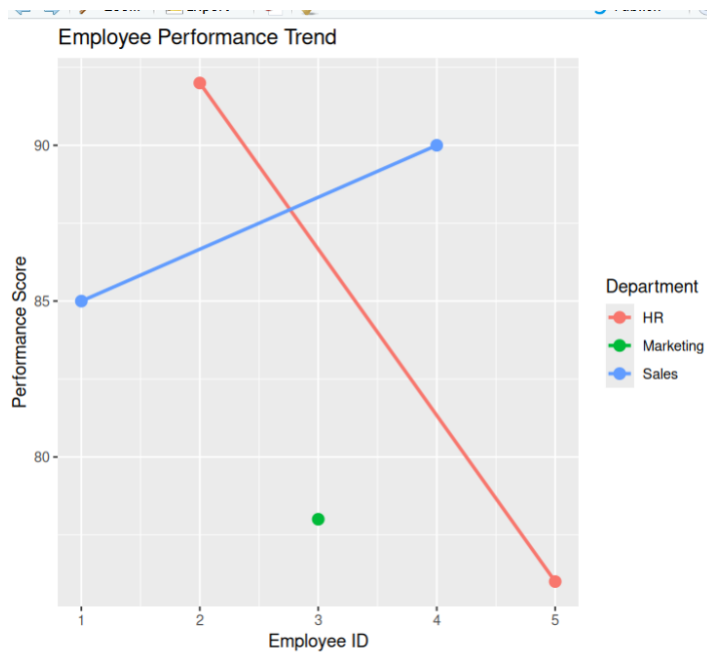
```
# Add both sheets to Dashboard.
```

```
# Enable filters for Department.
```

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Set 4: Product Inventory Management (R Code)

R

Load library

```
library(ggplot2)
```

Dataset

```
inventory <- data.frame(
```

```
  Product_ID = c(1,2,3,4,5),
```

```
  Product_Name = c("Product A","Product B","Product C","Product D","Product E"),
```

```
  Quantity_Available = c(250,175,300,200,220),
```

```
  Category =  
c("Electronics","Electronics","Furniture","Furniture","Accessories"),
```

```
  Price = c(120,150,100,180,140)
```

```
)
```

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```
# -----
```

```
# 1. Bar Chart - Quantity of Each Product
```

```
# -----
```

```
ggplot(inventory,  
  aes(x = Product_Name,  
    y = Quantity_Available,  
    fill = Product_Name)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Product Inventory Quantity",  
  x = "Product Name",  
  y = "Quantity Available"  
)
```

```
# -----
```

```
# 2. Stacked Bar Chart - Product Quantity by Category
```

```
# -----
```

```
ggplot(inventory,  
  aes(x = Category,  
    y = Quantity_Available,  
    fill = Product_Name)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Inventory Quantity by Category",  
  x = "Category",  
  y = "Quantity Available",
```

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```
    fill = "Product"
  )

# -----
# 3. Scatter Plot - Price vs Quantity
# -----

ggplot(inventory,
       aes(x = Price,
           y = Quantity_Available,
           color = Product_Name)) +
  geom_point(size = 4) +
  labs(
    title = "Product Price vs Quantity Available",
    x = "Price ($)",
    y = "Quantity Available"
  )

# -----
# Findings:
# Products with lower prices tend to have higher
# inventory quantities, while higher-priced
# products tend to have lower stock levels.
# -----

# -----
# 4. Tableau Dashboard
# -----
```

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1. Import dataset into Tableau.

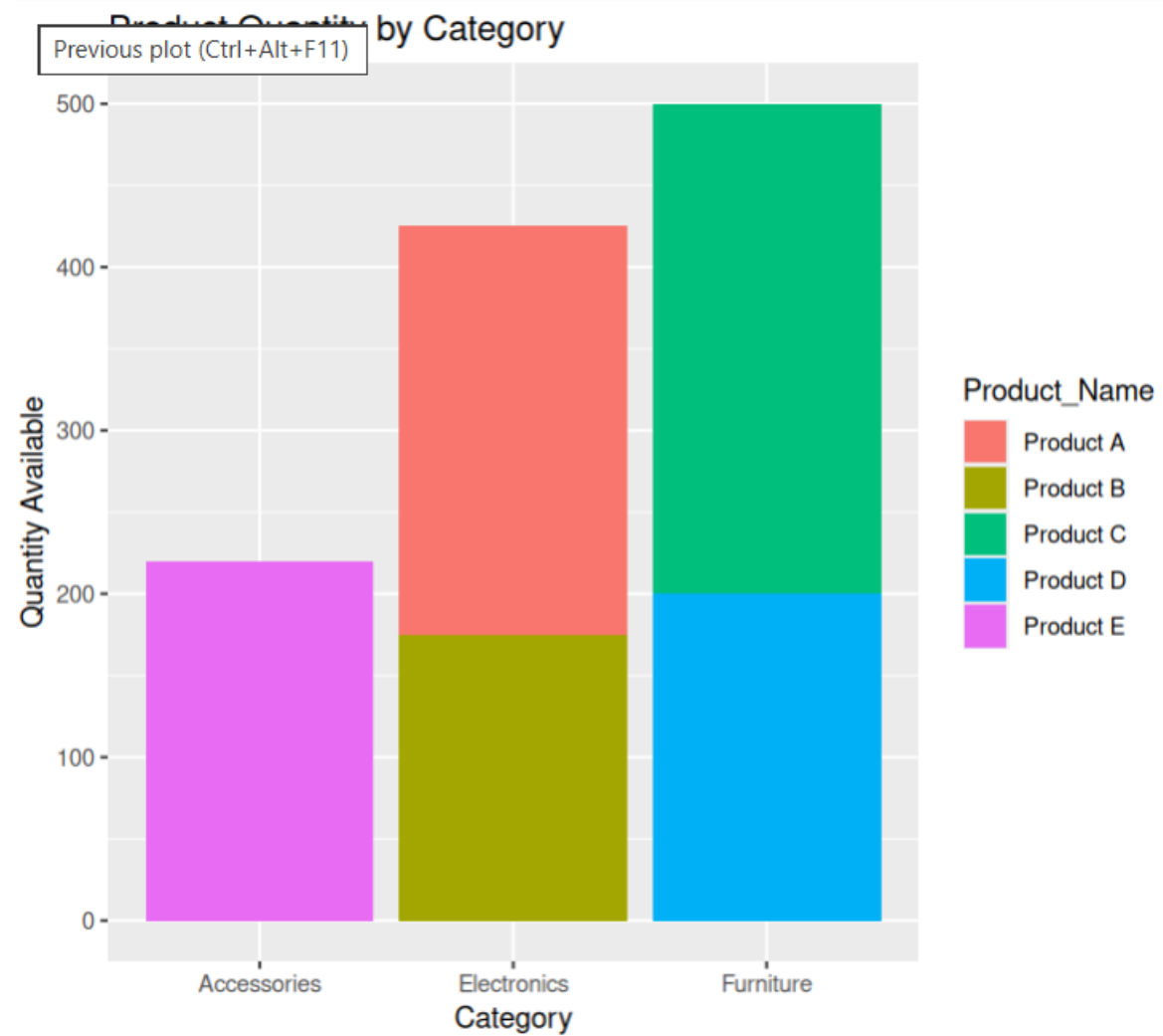
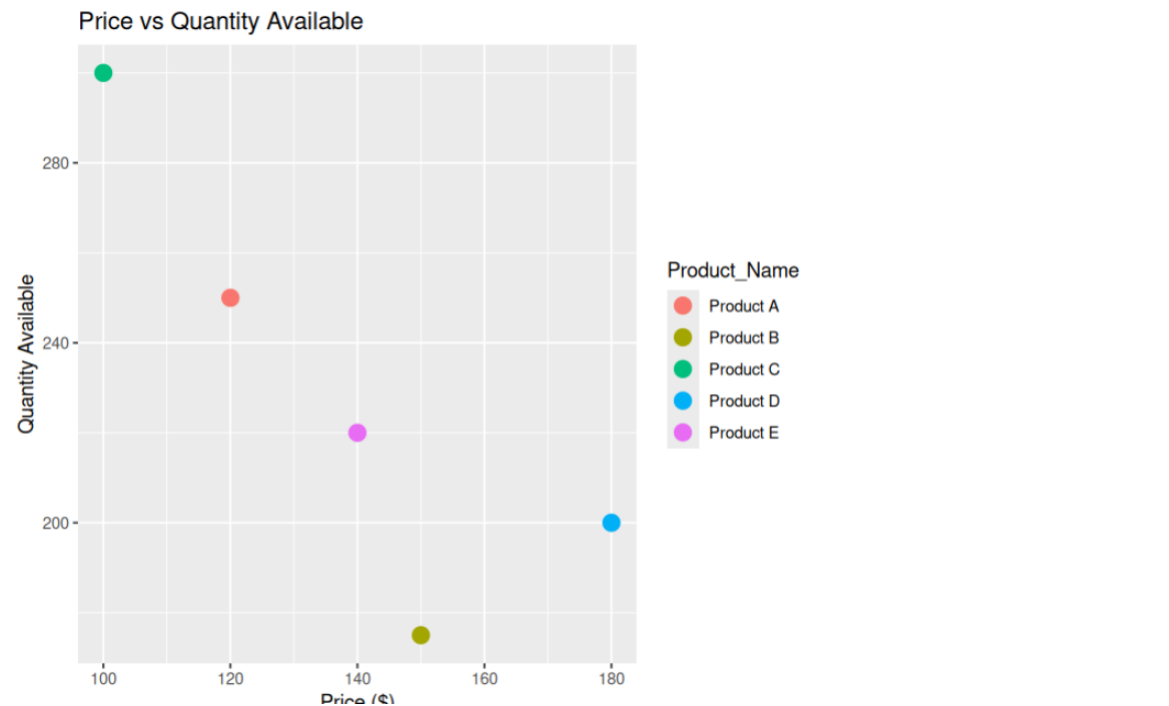
2. Create Sheet 1: Bar Chart.

3. Create Sheet 2: Stacked Bar Chart.

4. Add both sheets to a dashboard.

5. Add Category filter for interaction.

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Set 5: Website Analytics — Full R Code

Copy and paste this entire code into **Posit Cloud** / **RStudio** and run it.

R

Install packages (run once)

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

Load libraries

```
library(ggplot2)
```

```
library(reshape2)
```

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```
# -----
```

```
# Dataset: Website Traffic
```

```
# -----
```

```
website <- data.frame(  
  Date = as.Date(c(  
    "2023-01-01",  
    "2023-01-02",  
    "2023-01-03",  
    "2023-01-04",  
    "2023-01-05"  
  )),  
  Page_Views = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

```
print(website)
```

```
# =====
```

```
# 1. Line Chart - Daily Page Views
```

```
# =====
```

```
ggplot(website, aes(x = Date, y = Page_Views)) +  
  geom_line(color = "blue", linewidth = 1) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Page Views Trend",  
    x = "Date",  
    y = "Page Views"
```

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)

```
# =====
```

```
# 2. Bar Chart - Click Through Rate
```

```
# =====
```

```
ggplot(website, aes(x = Date, y = CTR)) +  
  geom_bar(stat = "identity", fill = "skyblue") +  
  labs(  
    title = "Click Through Rate by Date",  
    x = "Date",  
    y = "CTR (%)"  
  )
```

```
# =====
```

```
# 3. Stacked Area Chart
```

```
# =====
```

```
# Sample interaction data
```

```
likes <- c(100, 120, 90, 110, 130)
```

```
shares <- c(50, 60, 40, 55, 70)
```

```
comments <- c(20, 25, 18, 22, 28)
```

```
interaction <- data.frame(  
  Date = website$Date,  
  Likes = likes,  
  Shares = shares,  
  Comments = comments
```

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```
)
```

```
# Convert wide data to long format
```

```
interaction_long <- melt(
```

```
  interaction,
```

```
  id.vars = "Date"
```

```
)
```

```
# Stacked Area Chart
```

```
ggplot(interaction_long,
```

```
  aes(
```

```
    x = Date,
```

```
    y = value,
```

```
    fill = variable
```

```
  )) +
```

```
geom_area() +
```

```
labs(
```

```
  title = "User Interactions Over Time",
```

```
  x = "Date",
```

```
  y = "Count",
```

```
  fill = "Interaction Type"
```

```
)
```

```
# =====
```

```
# 4. Tableau Dashboard Instructions
```

```
# =====
```

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Import the dataset into Tableau

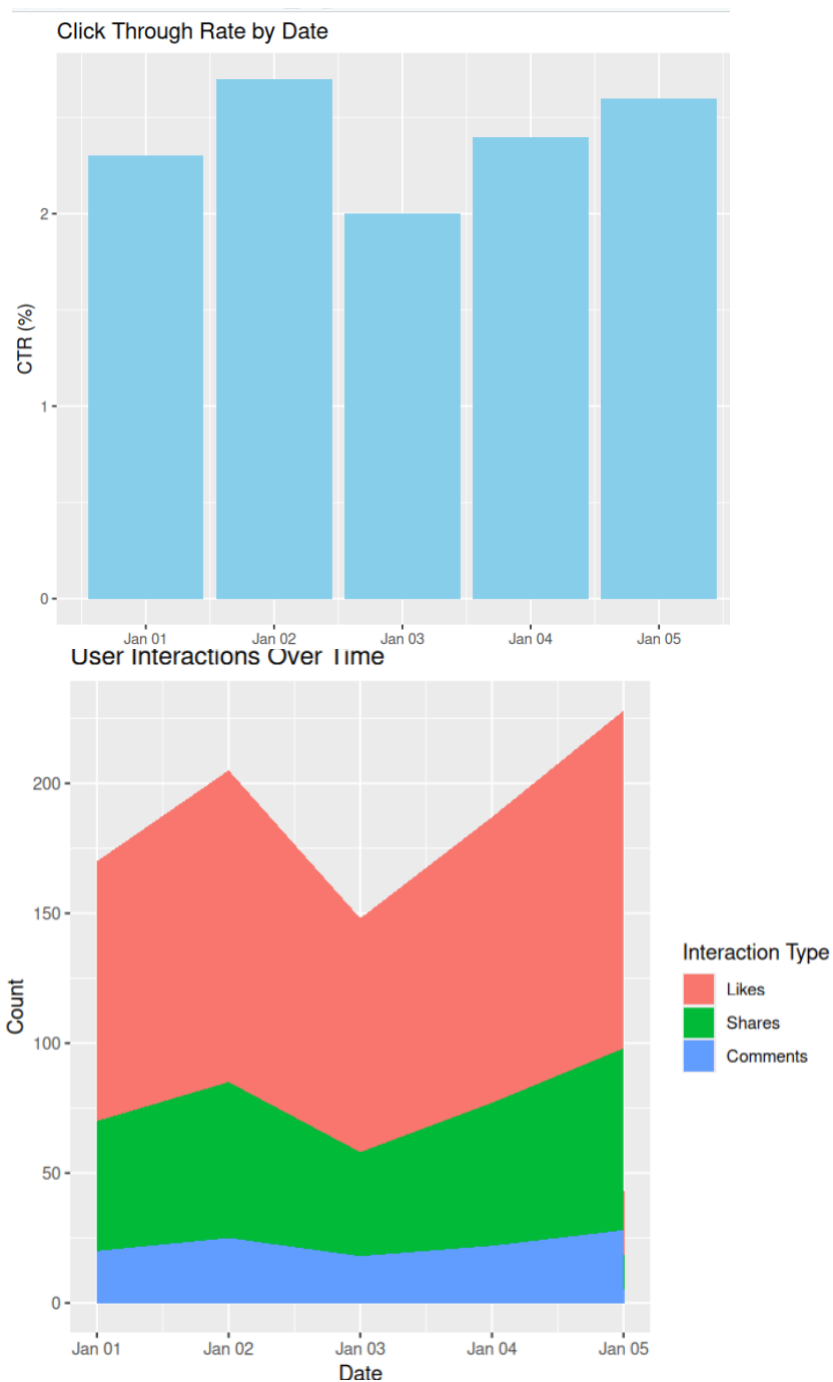
Create:

Sheet 1 -> Line Chart (Page Views)

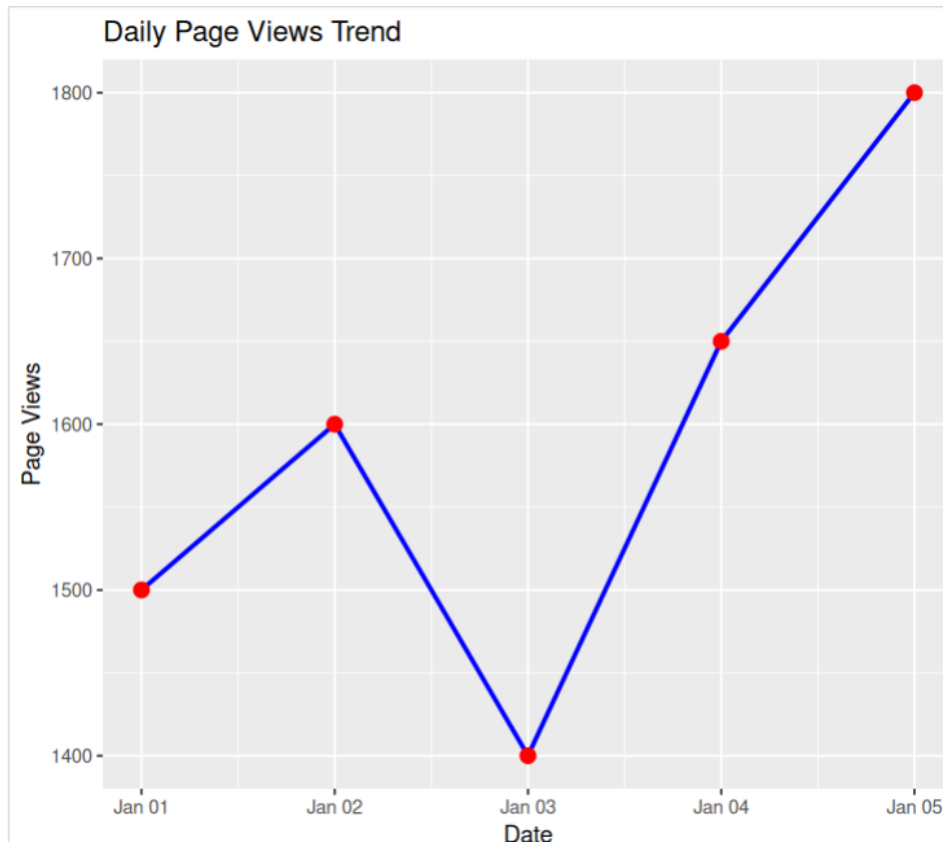
Sheet 2 -> Bar Chart (CTR)

Add both sheets to Dashboard

Add Date filter for interaction



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Set 6: Product Sales Analysis — Full R Code

Copy and paste this code into **Posit Cloud** / **RStudio**.

R

Install packages (run once)

```
install.packages("ggplot2")
```

```
install.packages("reshape2")
```

Load libraries

```
library(ggplot2)
```

```
library(reshape2)
```

-----

Dataset: Monthly Product Sales

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```
# -----
```

```
sales <- data.frame(  
  Product_ID = c(1,2,3),  
  Product_Name = c("Product A","Product B","Product C"),  
  January_Sales = c(2000,1500,1200),  
  February_Sales = c(2200,1800,1400),  
  March_Sales = c(2400,1600,1100)  
)
```

```
print(sales)
```

```
# Convert data to long format
```

```
sales_long <- melt(  
  sales,  
  id.vars = c("Product_ID","Product_Name"),  
  variable.name = "Month",  
  value.name = "Sales"  
)
```

```
# =====
```

```
# 1. Grouped Bar Chart
```

```
# =====
```

```
ggplot(  
  sales_long,  
  aes(  
    x = Product_Name,  
    y = Sales,
```


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```
    fill = Month
  )
) +
  geom_bar(
    stat = "identity",
    position = "dodge"
  ) +
  labs(
    title = "Quarterly Product Sales",
    x = "Product Name",
    y = "Sales Amount",
    fill = "Month"
  )

# =====

# 2. Stacked Area Chart

# =====

ggplot(
  sales_long,
  aes(
    x = Month,
    y = Sales,
    fill = Product_Name,
    group = Product_Name
  )
) +
  geom_area() +
```

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```
labs(  
  title = "Overall Sales Trend",  
  x = "Month",  
  y = "Sales",  
  fill = "Product"  
)
```

```
# =====
```

```
# 3. Monthly Sales Table
```

```
# =====
```

```
print(sales)
```

```
# =====
```

```
# 4. Tableau Dashboard Instructions
```

```
# =====
```

```
# 1. Import dataset into Tableau.
```

```
# 2. Create Grouped Bar Chart.
```

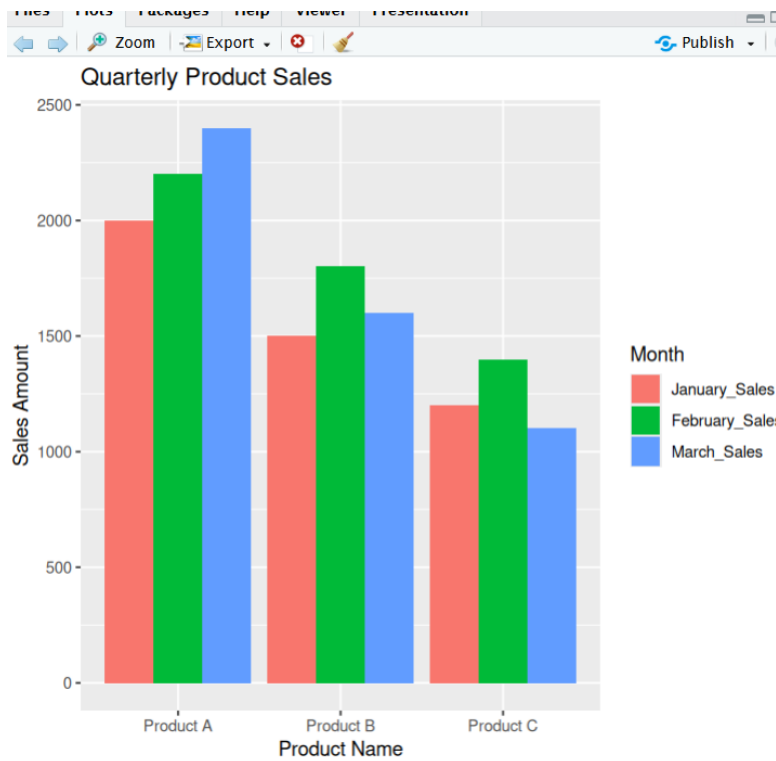
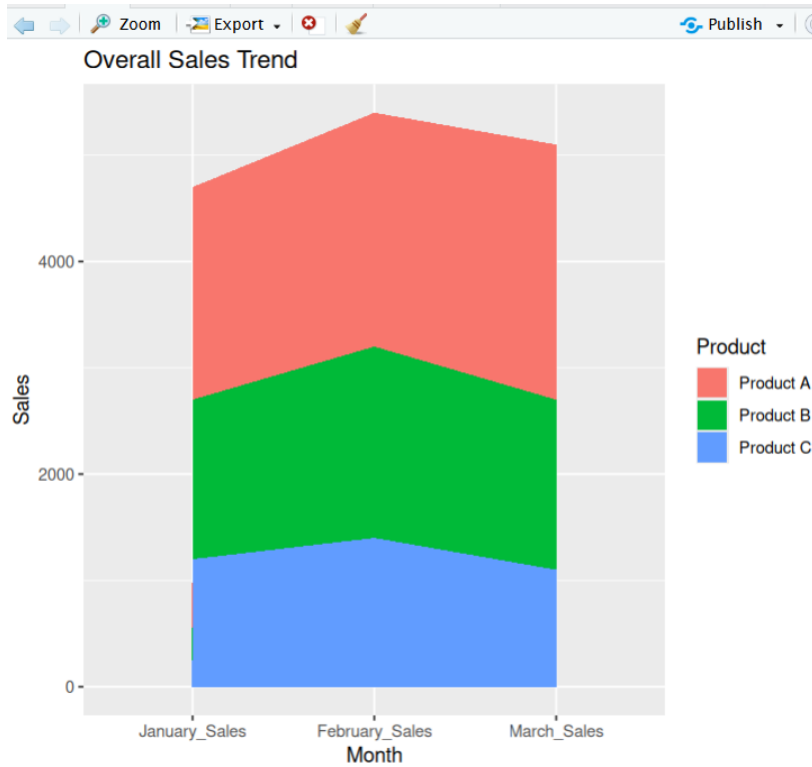
```
# 3. Create Stacked Area Chart.
```

```
# 4. Create Table View.
```

```
# 5. Add all three sheets to Dashboard.
```

```
# 6. Add Product filter for interaction.
```

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Set 7: Customer Demographics Analysis — Full R Code

Copy and paste this code into **Posit Cloud** / **RStudio**.

R

Install package (run once)

```
install.packages("ggplot2")
```

Load library

```
library(ggplot2)
```

-----

Dataset: Customer Demographics

-----

```
customer <- data.frame(  
  Customer_ID = c(1,2,3),  
  Age = c(28,35,42),  
  Gender = c("Female","Male","Female"),  
  Income = c(50000,60000,75000)  
)
```

```
print(customer)
```

=====

1. Bar Chart - Customer Ages

=====

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```
ggplot(customer,  
  aes(x = factor(Age))) +  
  geom_bar(fill = "skyblue") +  
  labs(  
    title = "Distribution of Customer Ages",  
    x = "Age",  
    y = "Number of Customers"  
  )
```

```
# =====
```

```
# 2. Pie Chart - Gender Distribution
```

```
# =====
```

```
gender_count <- table(customer$Gender)
```

```
pie(  
  gender_count,  
  labels = names(gender_count),  
  main = "Customer Gender Distribution"  
)
```

```
# =====
```

```
# 3. Table - Customer Information
```

```
# =====
```

```
print(customer)
```

```
# =====
```

```
# 4. Tableau Dashboard Instructions
```

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=====

Step 1: Import dataset into Tableau

Step 2: Create Bar Chart for Age Distribution

Step 3: Create Pie Chart for Gender Distribution

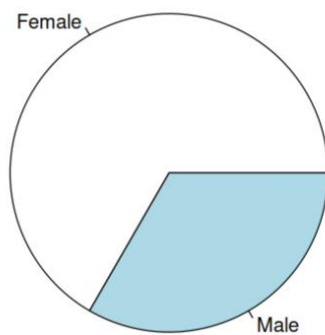
Step 4: Create Table for Customer Details

Step 5: Add all three sheets to Dashboard

Step 6: Add Gender filter for interaction

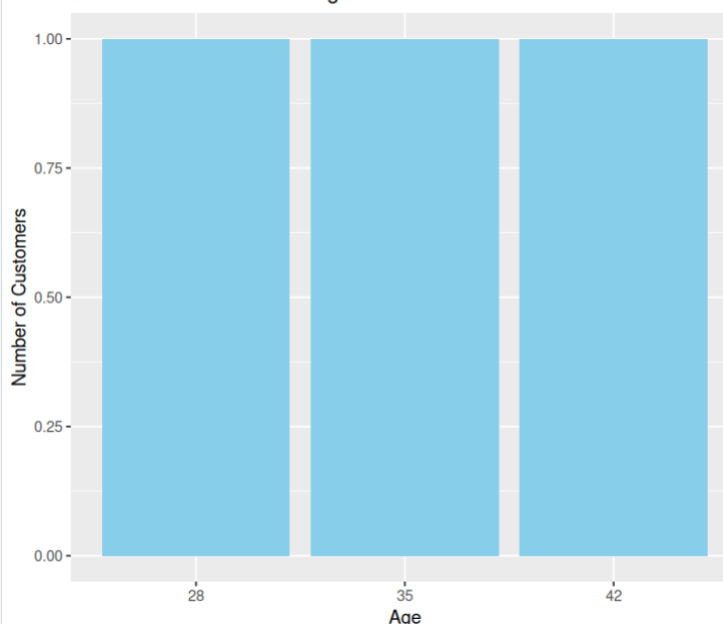
Zoom Export Publish

Customer Gender Distribution



Zoom Export Publish

Distribution of Customer Ages



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